

SMART MULTI-SPECIALITY HOSPITAL NETWORK

Edge & Core Router IP Configuration — WAN Links

Part 1: ISP (R0) <-> Edge Router Links — 200.1.1.0/24

Site	Router	Interface	IP Address	Subnet Mask / Prefix
Main Hospital HQ	R0	Gi0/0.101	200.1.1.2	/30
Main Hospital HQ	R1 (Edge)	Gi0/0	200.1.1.1	/30
Diagnostic Center	R0	Gi0/0.104	200.1.1.6	/30
Diagnostic Center	R4 (Edge)	Gi0/0	200.1.1.5	/30
Emergency Clinic	R0	Gi0/0.106	200.1.1.10	/30
Emergency Clinic	R6 (Edge)	Gi0/0	200.1.1.9	/30
Telemedicine	R0	Gi0/0.109	200.1.1.14	/30
Telemedicine	R9 (Edge)	Gi0/0	200.1.1.13	/30
Maternity Hospital	R0	Gi0/0.111	200.1.1.18	/30
Maternity Hospital	R11 (Edge)	Gi0/0	200.1.1.17	/30
Children Care Hospital	R0	Gi0/0.113	200.1.1.22	/30
Children Care Hospital	R13 (Edge)	Gi0/0	200.1.1.21	/30

Part 2: Edge Router <-> Core Router Links — 10.255.0.0/24

Site	Edge Router (Interface)	Edge IP	Core Router (Interface)	Core IP	Subnet
Main Hospital HQ	R1 (Gi0/1)	10.255.0.1	R2 (Gi0/0)	10.255.0.2	/30
Diagnostic Center	R4 (Gi0/1)	10.255.0.5	R5 (Gi0/0)	10.255.0.6	/30
Emergency Clinic	R6 (Gi0/1)	10.255.0.9	R7 (Gi0/0)	10.255.0.10	/30
Telemedicine	R9 (Gi0/1)	10.255.0.13	R10 (Gi0/0)	10.255.0.14	/30
Maternity Hospital	R11 (Gi0/1)	10.255.0.17	R12 (Gi0/0)	10.255.0.18	/30

Site	Edge Router (Interface)	Edge IP	Core Router (Interface)	Core IP	Subnet
Children Care Hospital	R13 (Gi0/1)	10.255.0.21	R14 (Gi0/0)	10.255.0.22	/30

CORE ROUTER — VLAN SUB-INTERFACE IP CONFIGURATION

Router-on-a-stick sub-interfaces per site (dot1Q encapsulation)

Site (Core Router)	VLAN	Department	Sub-Interface	IP Address	Prefix
Main Hospital HQ (R2)	10	DMZ	Gi0/1.10	10.0.1.161	/28
Main Hospital HQ (R2)	20	Doctors & ICU	Gi0/1.20	10.0.1.65	/26
Main Hospital HQ (R2)	30	Billing & Admin	Gi0/1.30	10.0.1.1	/26
Main Hospital HQ (R2)	50	Guest Wi-Fi	Gi0/1.50	10.0.0.1	/24
Diagnostic Center (R5)	100	Laboratory	Gi0/1.100	10.1.0.1	/26
Diagnostic Center (R5)	110	Admin	Gi0/1.110	10.1.0.65	/27
Diagnostic Center (R5)	120	Staff	Gi0/1.120	10.1.0.97	/28
Emergency Clinic (R7)	200	Emergency Ward	Gi0/1.200	10.2.0.1	/26
Emergency Clinic (R7)	210	Admin	Gi0/1.210	10.2.0.65	/27
Emergency Clinic (R7)	220	IP Phone	Gi0/1.220	10.2.0.97	/28
Telemedicine (R10)	300	Consultation	Gi0/1.300	10.3.0.1	/26
Telemedicine (R10)	310	Admin	Gi0/1.310	10.3.0.65	/27
Telemedicine (R10)	320	Wi-Fi	Gi0/1.320	10.3.0.129	/27
Maternity Hospital (R12)	400	Maternity Ward	Gi0/1.400	10.4.0.1	/26
Maternity Hospital (R12)	410	Admin	Gi0/1.410	10.4.0.65	/27
Maternity Hospital (R12)	420	Wi-Fi	Gi0/1.420	10.4.0.129	/27
Children Care Hospital (R14)	500	Pediatric Ward	Gi0/1.500	10.5.0.1	/26
Children Care Hospital (R14)	510	Admin	Gi0/1.510	10.5.0.65	/27
Children Care Hospital (R14)	520	Wi-Fi	Gi0/1.520	10.5.0.129	/27

NOTES & VERIFICATION

- This document reflects all IP addressing confirmed through this conversation's CLI configs, screenshots, and command outputs (show ip interface brief, show run, etc.).
- R0 (ISP router) uses sub-interfaces on Gi0/0 (dot1Q trunked via the ISP-side switch) to reach each Edge router over its own dedicated /30 subnet.
- Edge routers (R1, R4, R6, R9, R11, R13) each use a plain Gi0/0 (toward R0) and Gi0/1 (toward their Core router) — no sub-interfaces on the Edge side.
- Core routers (R2, R5, R7, R10, R12, R14) use Gi0/0 toward their Edge router, and Gi0/1 sub-interfaces (dot1Q per VLAN) toward their site's Layer 2 switch trunk.
- R11/R12 (Maternity) later had VLAN 420 (Wi-Fi) added, matching the pattern used at Diagnostic Center, Telemedicine, and Children Care Hospital.
- Router 9 and Router 11 previously had EIGRP network statements copied from R12/R13, which was corrected in a separate fix (see EIGRP Fix Report).

Recommended verification: Run `show ip interface brief` on each router to confirm these addresses are still current, since manual re-configuration during troubleshooting may have changed individual interfaces since they were last confirmed in this conversation.